

Topological Board Game - Quick Start Guide

Language: English

This guide is for players opening the Steam version for the first time. It explains what to click, how to start a game, and what the main modes mean.

1. Start The Game

- Open Topological Board Game from Steam.
- Choose a game family: Chess, Go, Reversi, Jump / Chinese Checkers, Hex, Life World, or Labs.
- Choose a dimension or mode: 2D, 3D, 4D, 2+1D, or 3+1D.
- Choose Local, Online, or Robot where available.
- Press New Game or Start.

2. Basic Controls

- Left click or tap a board site to select, place, or move.
- Drag to rotate 3D boards when the page supports a rotatable view.
- Use mouse wheel or pinch gestures to zoom.
- Use Reset Camera if the board moves out of view.
- Use Home to return to the main launcher.
- Use the globe icon to switch English and Traditional Chinese.

3. Game Families

Chess

- Move one piece at a time according to standard chess piece movement.
- Protect your king and try to checkmate the opponent king.
- Topological boards change which squares are adjacent or connected, but the chess rule is still the chess rule.

Go

- Place black and white stones on board intersections.
- Stones with no liberties are captured.
- Pass when you do not want to play.
- At the end, territory and captures are counted to decide the winner.

Reversi

- Place a disc so that at least one straight connected line of opponent discs is bracketed.
- Bracketed discs flip to your color.

- The player with more discs at the end wins.

Jump / Chinese Checkers

- Move or jump pieces along the board graph.
- Reach the target camp before the opponent.
- Topology changes the connection graph, so legal jumps follow the shown links.

Hex

- Blue and Orange place cells on empty Hex board units.
- Blue connects its two target sides.
- Orange connects its two target sides.
- There is no movement, flipping, capture, or passing by default.

4. Dimensions And Topology

- 2D: flat board-game play.
- 3D: rotatable boards, surfaces, volumes, or embedded boards.
- 4D: higher-dimensional boards shown through slices or projection controls.
- 2+1D and 3+1D: board games with a time layer.
- Boundary and topology options change how edges connect, such as open boundary, cylinder, torus, Mobius strip, Klein bottle, sphere, or projective-style boundary.

5. Time Modes In +1D Games

- Future Mode: choose a move now; it resolves after a delay.
- Past Mode: rewrite a recent resolved move inside a fixed window, then replay the timeline.
- Past+Future Mode: combine scheduling future moves and rewriting recent past moves.
- Age and Go Time Period are separate optional modes where available.
- Ordinary 2D, 3D, and 4D games are normal board games and do not need time modes.

6. Online Play

- Use Find Match or Room controls where available.
- If sign-in is required, use the account panel.
- If password login says the sign-in method is not enabled, enable Email/Password in Firebase Authentication for the project.

7. Life World

- Life World is a cellular automata and artificial-life mode.
- Use Start, Step, Reset, Random seed, drawing, erasing, species selection, and pattern tools.

- Research controls include topology, lattice, B/S rules, observables, phase scans, and topology comparisons.

8. Labs

- Labs is the scientific research area.
- It is for topology, graph dynamics, spin systems, quantum information, topological matter, and field-theory style experiments.
- Labs is not ordinary gameplay; it focuses on reproducible experiments, observables, export, and comparison.

9. Quick Troubleshooting

- If the board is off screen, use Reset Camera or reduce zoom.
- If online play does not connect, check internet, Firebase config, and room settings.
- If a 3D board feels slow, reduce board size or use a simpler lattice.
- If language is wrong, use the globe icon and reload.

10. Steam Manual Files

This PDF is a quick reference. For complete rules, use the Full Manual PDF. For installation and support, use the README and Troubleshooting PDF.